Assignment 2 Report | Geostrophic Transport and Relating AMOC Series

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- **Course** : 63-731 Data Analysis in Physical Oceanography

1. Introduction

1.1 Aim

- Compute the 26°N UMO geostrophic transport, compare it with the official UMO product, and compute the correlation coefficient between calcualted UMO transport and the official UMO transport timeseries.

- Calculate and illustrate the monthly climatology of both the 26°N MOC and UMO transport series. Plot the raw and deseasonalised figures for both timeseries.

- Perform an autocorrelation with T* marked, fit a linear trend for both the 26°N MOC and UMO transports series, and find the significance based on N_eff.

- Match 2 series, 26°N MOC and 47°N MOC to show a cross-correlation analysis for after removing their seasonal cycles. Draw scatter plots at the peak lag for both raw and deseasonalised data. Evaluate the significance.

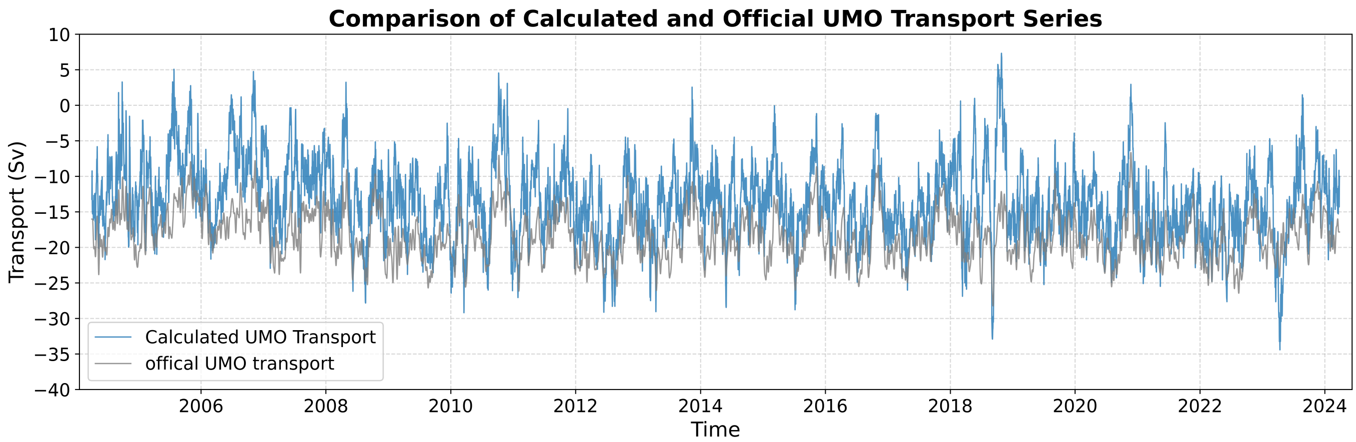
- Perform a depth sensitivity analysis for the 26°N UMO transport to assess the effect of different max depths on the integrated transport values.

1.2 Dataset

1. `ts_gridded.nc`. Link: rapid.ac.uk/data/gridded-mooring-data
    - The gridded mooring data contains 5 vertical profiles, and all profiles comtain timeseries of 12-hourly temperature and salinity data gridded onto 20m intervals starting in Apr 2004.
2. `moc_transports.nc`. Link: rapid.ac.uk/data/integrated-transports
    - 12-hourly, 10-days low pass filtered transport timeseries from Apr 2004 to Mar 2024.

2. Results and Discussion

Part 1 Geostrophic Transport



This figure shows the comparison between the calculated UMO transport timeseries and official product. The 2 series shows very similar fluctuations, and obviously, the calclated one always higher than the official product, with the calcualted data occasionally appearing more aggressive.

Here is a table showing the results from the correlation analysis:

Parameter	**Value**
Valid Sample Points	14,599
Correlation (r)	0.762 ($p < 0.001$)
Variance Explained (R^2)	58.1%
Mean Bias (Calculated − Official)	+4.911 Sv
Root Mean Square Error (RMSE)	6.115 Sv

According to the table, the calculated data is higher than official data by an average of 4.911 Sv, while the correlation between them reaches 76.2% with p value < 0.001, suggesting that they share almost the same dynamics.

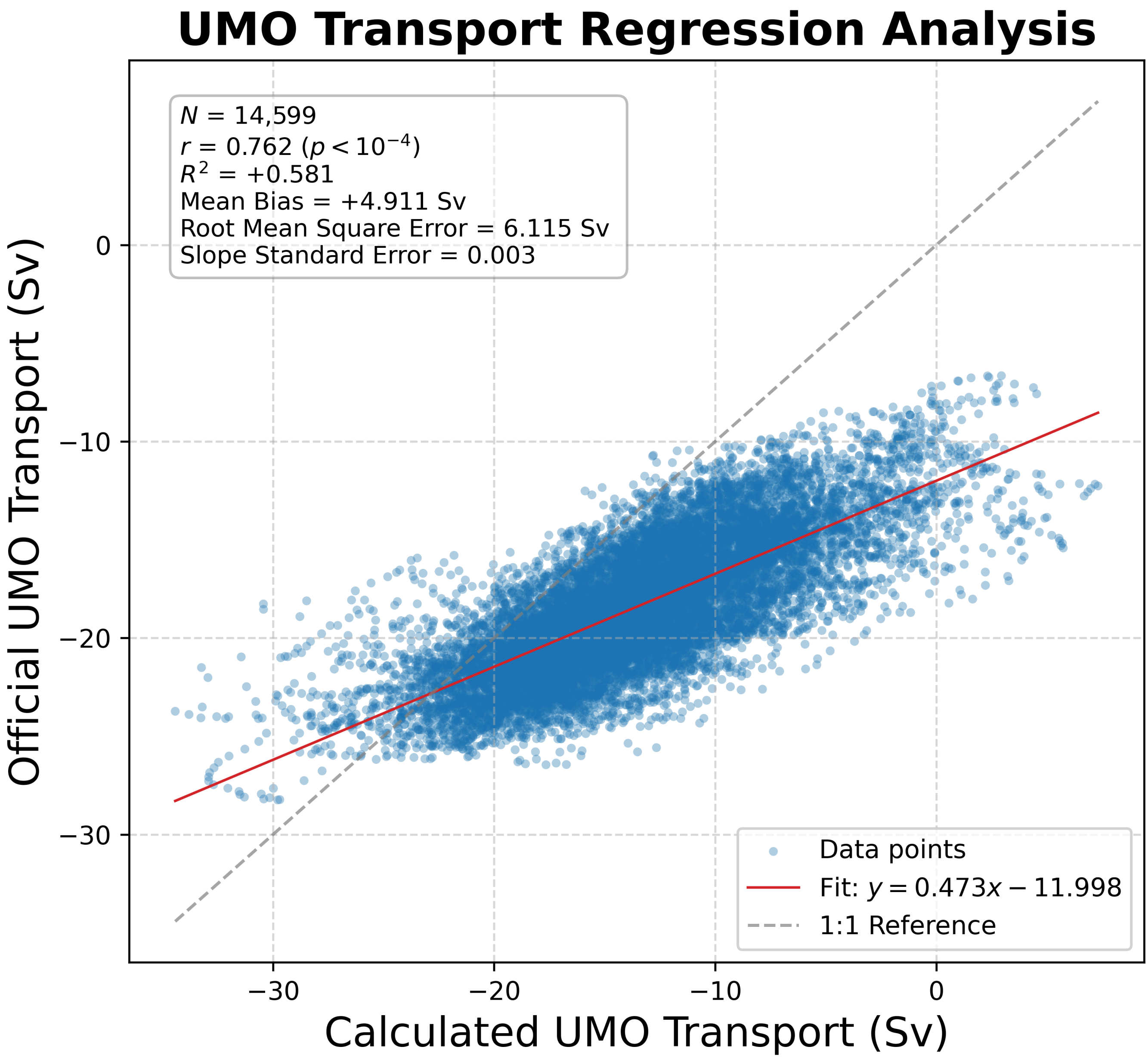
I checked the function `interior_geostrophic_transport`, which sets a fixed max depth of 1100m, while RAPID uses a time-varing value.

Besides, the function doesn't account for compensation transport, ignoring the mass-balance adjustment, while RAPID includes an external transport adjustment.

Overall, ignoring the compensation transport (which is always negative) is likely the main reason for the +4.911 Sv bias, and the fixed max depth of 1100m may explain the RMSE and aggressive fluctuations in the calculated UMO transport series.

Linear Regression fit:

$$\text{Official UMO} = 0.473 \times \text{Calculated UMO} - 11.998$$
$$\text{Slope Std Error} = 0.003$$

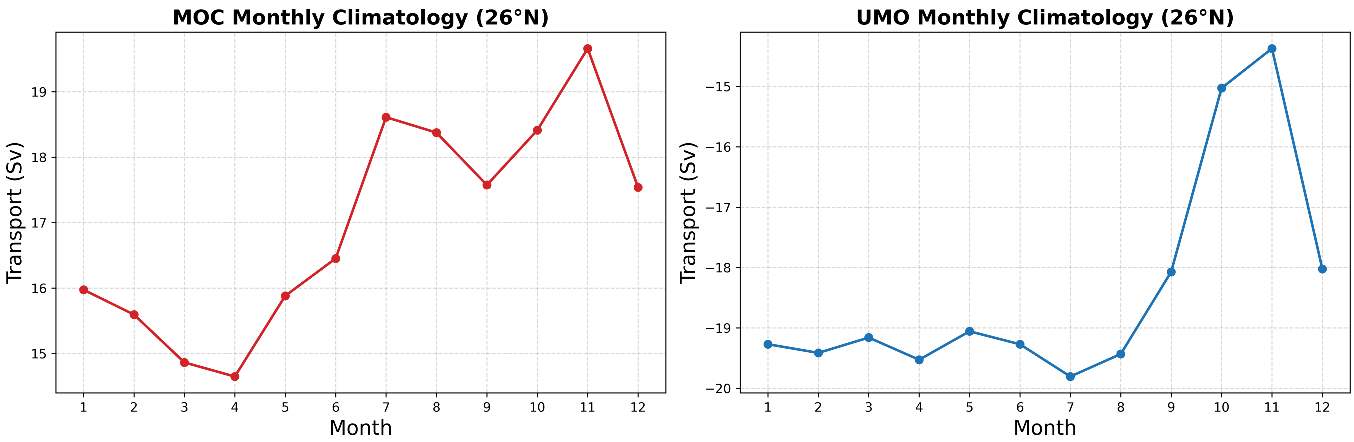


The second figure shows that the fit line shifts a lot from the 1:1 reference line. The slope of 0.47 indicates that the calculated data has roughly twice the amplitude of the official data. Besides, the intercept of -12 Sv matches the systematic bias discussed above.

> Q: Why the uppper mid-ocean is not a full-depth integral?
>
> A: At first, UMO happens in the upper layer between the surface and around 1100m depth, with 2 water masses(UNADW and LNADW) down below. Secondly, if we do the full-depth integral, the results will not tell us each layer's contribution separately.

Part 2 Evaluating time series

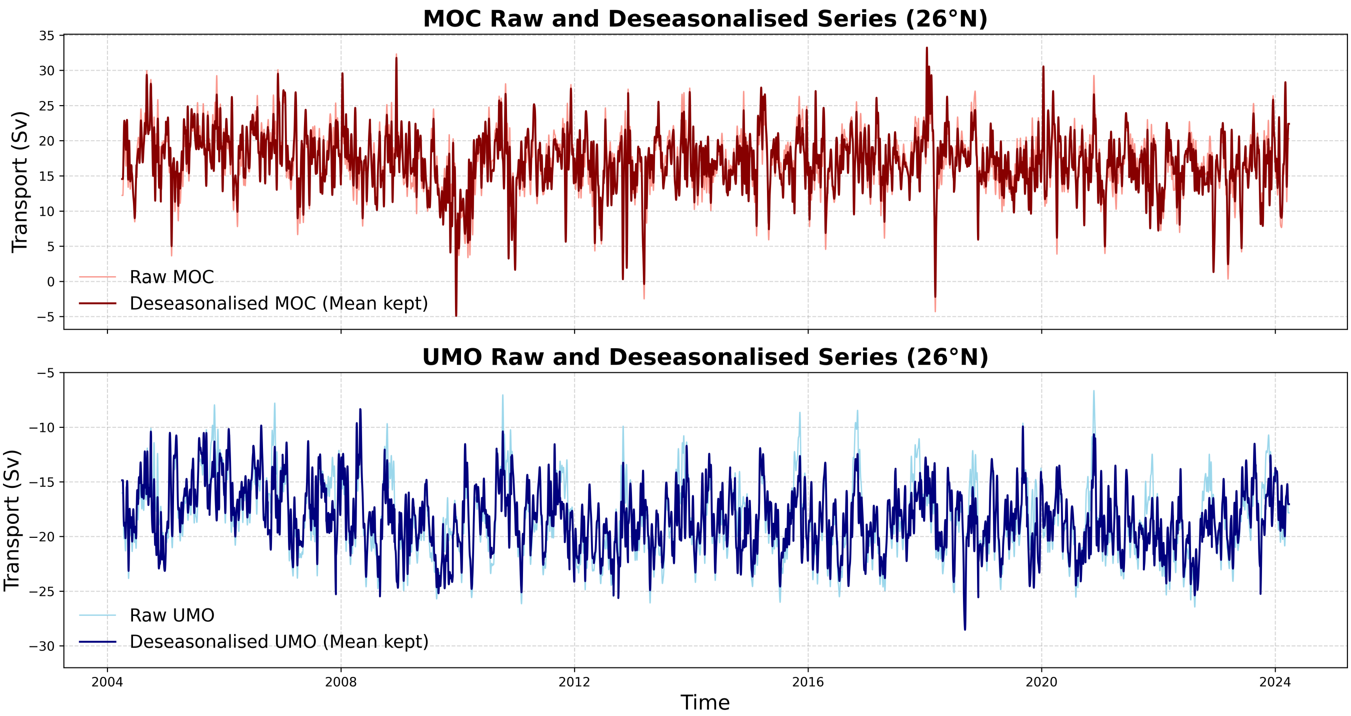
2A Seasonal cycle and trend (for both the 26°N MOC and UMO)



This climatology figures indicates that the MOC transport increases from April to June and decreases from Nov to March. Meanwhile, the UMO transport keeps stable from Jan to July, and decreases sharply starting from Aug.

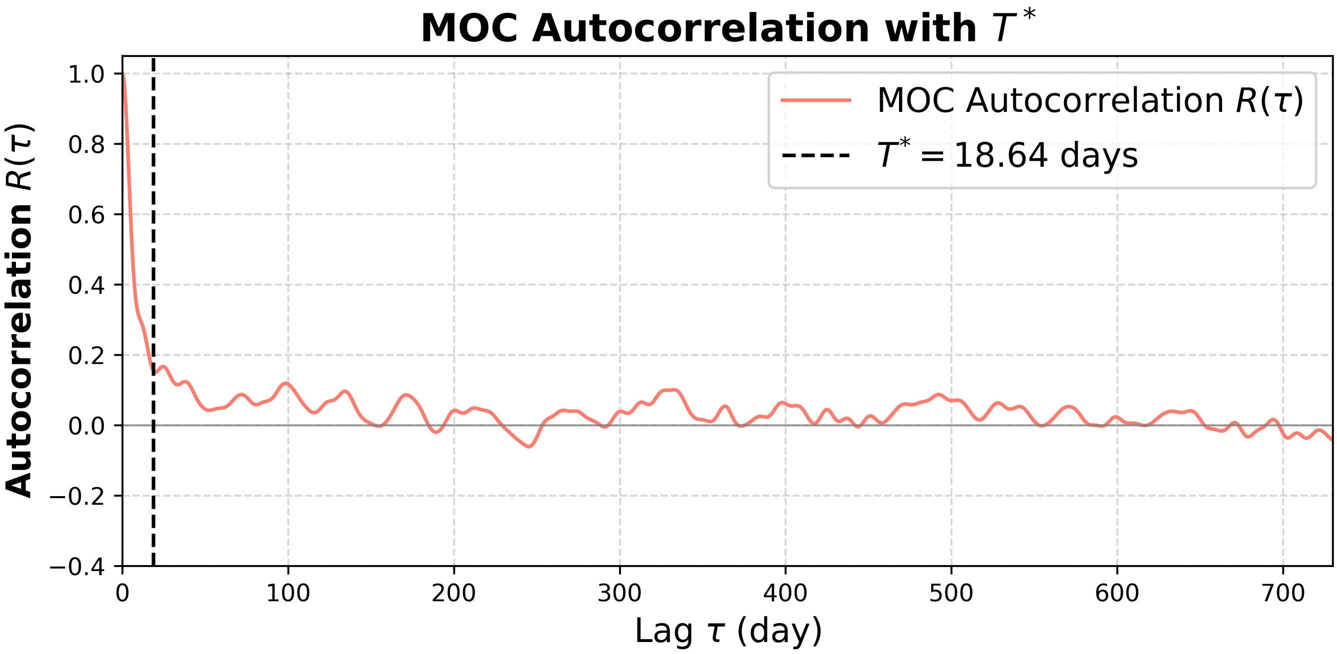
MOC and UMO transports have significant seasonal cycles. To calculate correlation timescales T* and the effective sample size N_{eff}, deseasonalisation is necessary.

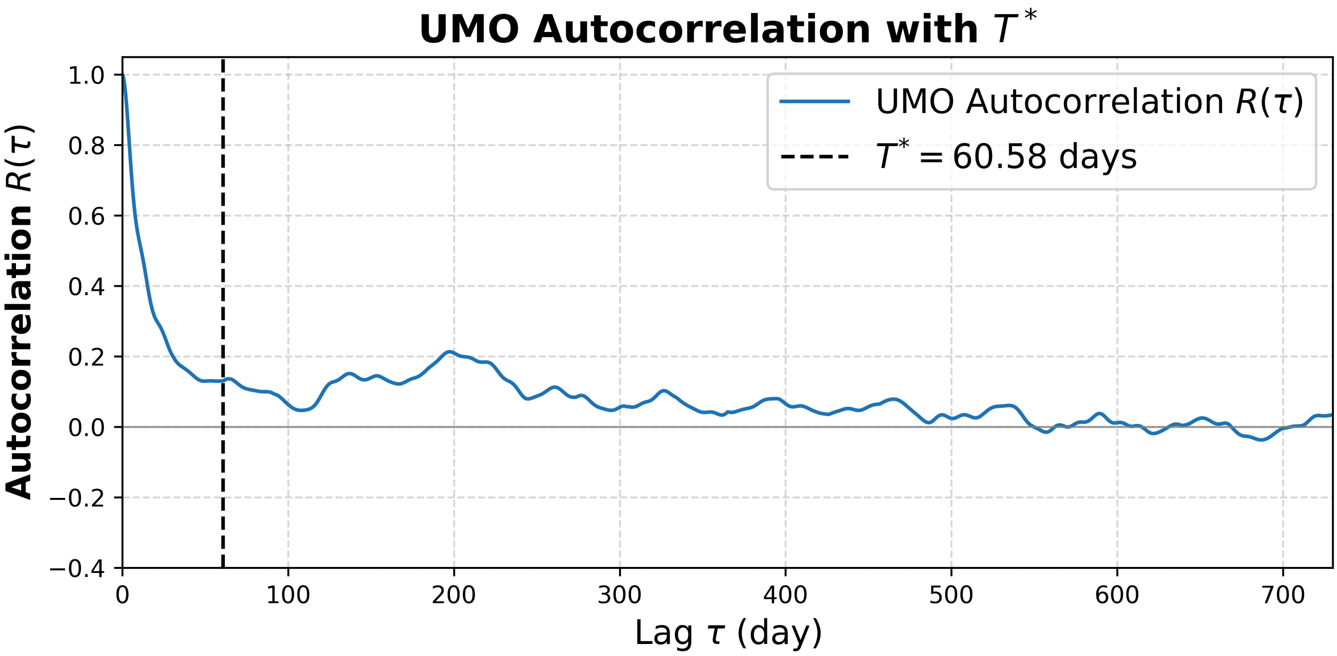
After filtering out the seasonal effects, the residual anomalies will provide a more reliable estimation of T*. (The overall mean is remained, but the `autocorr` function uses `d d-d.mean` to remove the mean when computing r and T*.)



The timeseries shows that the amplitude of the UMO series shrinks noticeably after deseasonalization.

The following figures depict the results of the autocorrelation for both the MOC and UMO timeseries.





Variable	Integral Timescale (T^*)
MOC	18.64 days
UMO	60.58 days

MOC has a much shorter T* relative to UMO according to the figures and the table above.

$$\text{UMO} = \text{Upper Geostrophic Transport} + \text{Compensation Transport}$$
$$\text{MOC} = \text{FC} + \text{Ek} + \text{UMO}$$

As the total sum of full-depth integrated transport, MOC manifests greater variability and high-frequency fluctuations. Therefore, its autocorrelation decays faster than that of UMO, which makes sense.

Here is a table showing the results from the linear regression trend analysis:

Parameter	MOC	UMO
Fitted Slope	$-0.093\ \mathrm{Sv/yr}$	$-0.109\ \mathrm{Sv/yr}$
Intercept	$+17.903\ \mathrm{Sv}$	$-17.274\ \mathrm{Sv}$
Naive Standard Error	$0.006\ \mathrm{Sv/yr}$	$0.004\ \mathrm{Sv/yr}$
Effective Standard Error	$0.047\ \mathrm{Sv/yr}$	$0.036\ \mathrm{Sv/yr}$
Effective Sample Size (N_{eff})	223.933	190.693
Slope / Std. Error	-1.960	-3.025
P-value	5.12×10^{-2}	2.83×10^{-3}
Significant (95%)	**No** $(p > 0.05)$	**Yes** $(p < 0.05)$

The average value of MOC decreases by 0.093 Sv for each year, while UMO decreases by 0.109 Sv annually. However, when we focus on the timeseries itself, it doesn't indicate a decreasing trend because strong noise masks it.

The project uses the `effective_dof` function to compute $N_{eff} \approx 224$ for MOC and $N_{eff} \approx 191$ for UMO. Compared with the original sample size $N = 14,599$, the effective sample size drops a lot due to autocorrelation.

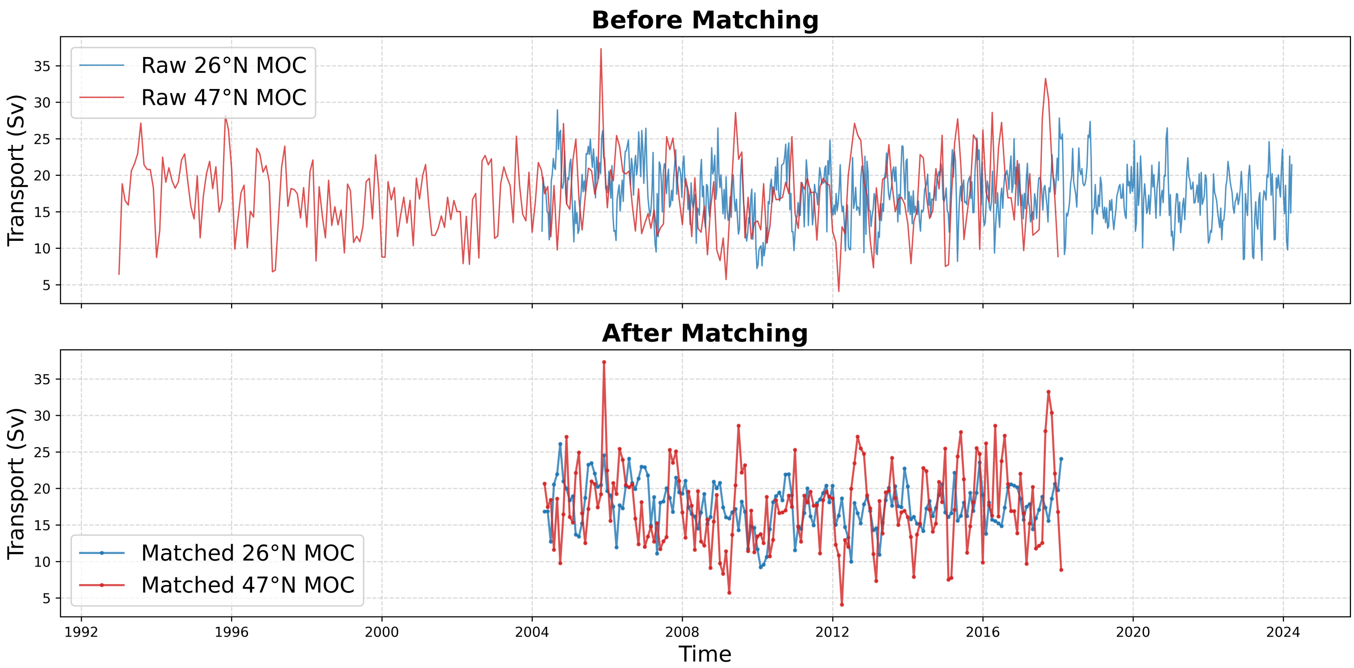
Additionally, the p value of MOC exceeds 0.05, which means the trend lacks significance and it shouldn't be treated as a concrete conclusion.

2B Cross-correlation of 26°N and 47°N MOC

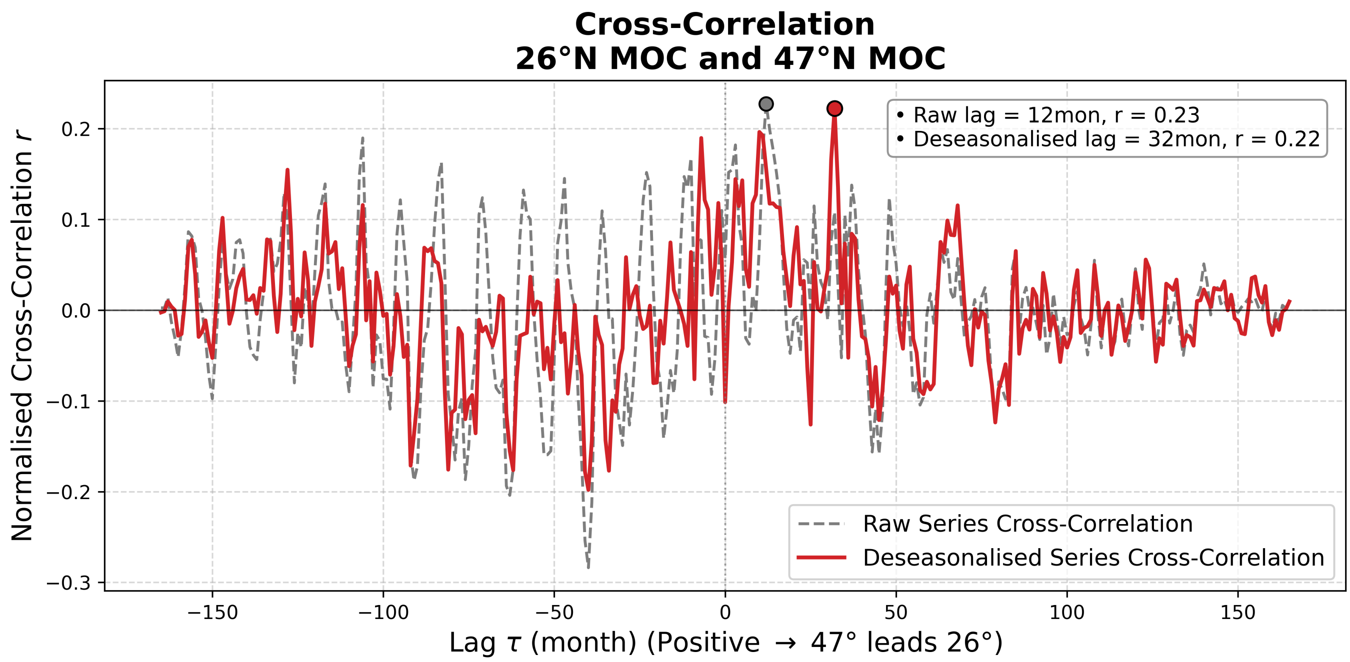
26°N MOC and 47°N MOC have different timespans and sampling frequences,

Variable	Start Date	End Date	Total Data Points (N)
26°N MOC	2004-04-06	2024-03-22	730
47°N MOC	1993-01-01	2018-01-01	301

therefore the two series need to be resampled on a common time grid before cross-correlation.



The following figure depict the cross-correlation results for the resampled 26°N and 47°N MOC timeseries.

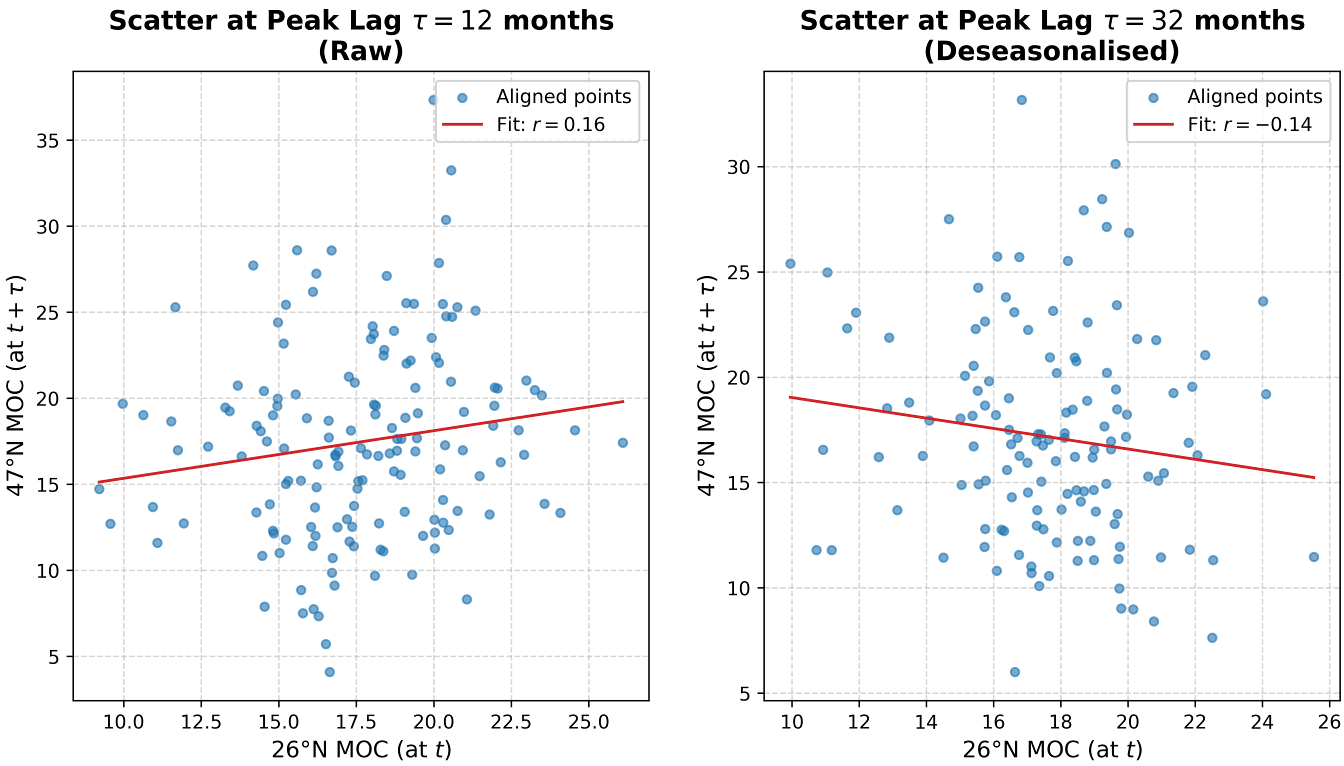


After removing the seasonal cycles, the positive cross-correlation peaks from $\tau = -100$ to $\tau = 0$ are significantly attenuatted. Without the effect of seasonal driving, the fluctuations seem more centralised.

Variable	Peak Lag (τ)	Correlation Coefficient (r)
Raw	12 months	0.227
Deseasonalised	32 months	0.222

Besides, the deseasonalised peak lag reaches 32 months, which is nearly 3 times that of the raw series. This implies that the underlying signal takes 32 months to be propagated from 47°N to 26°N.

If peak_lag > 0 (y leads x), we remove the last 32 months of x and the beginning 32 months of y, and generate a point-by-point scatter plot.



The two subplots reveal a weak relation between the 26°N and 47°N MOC timeseries. Futhermore, the linear trends have reversed directions, the raw series shows a positive slope, whereas the deseasonalised one demonstrates a negative slope overall.

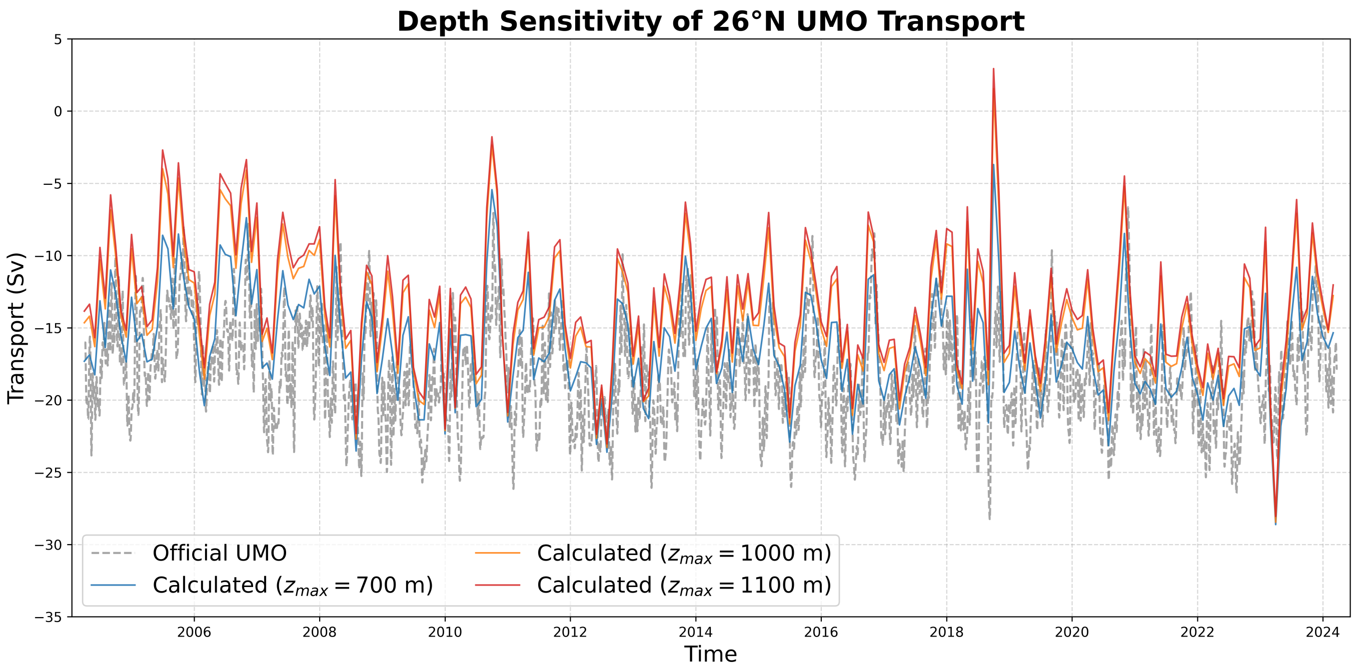
For the deseasonalised:

Variable	value	Description / Threshold
Effective Sample Size (N_{eff})	48.132	Accounts for autocorrelation in the series
95% Critical r	0.284	Significance threshold at the 95% confidence level
Peak r	0.222	Maximum cross-correlation coefficient
Significant	`False`	Peak r < Critical r

Clearly, the result lacks statistical significance. Therefore, it is only suggestive rather than a concrete conclusion.

Part 3 Depth Sensitivity Analysis

The analysis used the RAPID 26°N boundary hydrography to compute the UMO integral transport based on difference maximum depth.



Obviously, as the integration depth goes deeper, the calculated transport gets larger. However, the official product provides the smallest transport, because it accounts for the negative compensation transport as we discussed in Part 1.

Allowing the integration depth to vary over time is very essential, because the lower boundary of the UMO layer changes under the influence of AAIW.

From Oct to Jan, AAIW goes northward, deepening the northward upperlayer, requiring the max depth to be set around 1100m during the period. From Mar to Jun, as the influence of AAIW decays, the max depth decreases to around 700m.